Predicting US Housing Prices Using Machine Learning

A Comprehensive Analysis of County-Level Price Forecasting

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ABSTRACT

This research develops and evaluates machine learning models to predict median housing prices at the county level across the United States. Using the Redfin Housing Market dataset spanning 2012-2021 with **563,122** observations across 1,860 counties, implement a comprehensive data mining pipeline encompassing data quality remediation, systematic feature engineering creating 119 predictive variables, and rigorous temporal validation methodology. The final XGBoost gradient boosting model achieves exceptional performance with RMSE of \$4,574 and R^2 of 0.9983 on 2021 test data, substantially exceeding both project goals (RMSE < \$25,000, $R^2 > 0.85$). Feature importance analysis reveals that historical price lags at 1, 3, and 12 months collectively dominate predictive power accounting for 44 percent of total model importance, confirming that housing markets exhibit strong momentum over short horizons. This work demonstrates that proper temporal validation preventing data leakage, comprehensive feature engineering capturing market dynamics, and advanced gradient boosting methods enable highly accurate short-term housing price forecasting with practical applications for real estate investment, policy planning, and financial risk management.

1. INTRODUCTION

1.1 Background and Context

Residential real estate, valued at approximately $43.4 trillion in the United States as of 2021 (Zillow Research, 2021), represents one of the largest components of the national economy and the primary store of wealth for most American households. Housing price dynamics directly impact household financial security, economic stability, and access to homeownership. The complexity of housing markets stems from multiple interacting factors including supply and demand fundamentals, macroeconomic conditions, geographic variations, and seasonal patterns in buyer activity.

Accurate price forecasting serves diverse stakeholders. Homebuyers face critical timing decisions about market entry, balancing the risk of purchasing at peak prices against the opportunity cost of renting while prices appreciate. Real estate investors require reliable forecasts to allocate capital across geographic markets before broader recognition drives up valuations. Financial institutions depend on accurate valuations for mortgage underwriting, ongoing risk assessment, and loss mitigation strategies. Policymakers address housing affordability challenges and plan infrastructure investments informed by price trends and their implications for tax revenues and demographic stability.

Traditional valuation approaches rely on comparative market analysis by real estate professionals and formal appraisals for mortgage lending. While incorporating local expertise, these methods can be subjective in their selection of comparable properties, time-intensive requiring manual research, and inconsistent across different practitioners and markets. The proliferation of publicly available housing transaction data through platforms like Redfin and government sources, combined with advances in machine learning algorithms, presents an opportunity to develop systematic, data-driven forecasting models that complement traditional methods with empirical rigor, consistency, and scalability.

1.2 Research Objectives

This research addresses a fundamental question: Can machine learning models accurately predict future median housing prices at the county level using publicly available market data? I establish two quantitative performance targets representing practically useful accuracy. First, I aim to achieve Root Mean Squared Error below $25,000, approximately 13.6 percent of the national median home price, defining an error margin small enough to inform investment and policy decisions. Second, I target coefficient of determination exceeding 0.85, indicating the model explains at least 85 percent of price variance and provides substantial predictive power beyond naive baseline approaches.

Beyond these quantitative objectives, we address several analytical questions informing both model development and practical application. Which features most strongly influence county-level price movements? How do geographic and temporal factors interact in determining prices? Does XGBoost gradient boosting provide meaningful accuracy improvements over traditional regression and ensemble methods? What validation methodology ensures performance metrics honestly reflect real-world forecasting accuracy rather than merely fitting historical patterns through data leakage?

1.3 Contributions

This work makes three primary contributions. First, I provide empirical demonstration of the critical importance of temporal validation for time-series economic data, showing how random train-test splitting produces misleading performance metrics that fail to generalize. Second, establish comprehensive feature engineering protocols for housing market data, creating 119 predictive variables organized into temporal patterns, geographic hierarchies, and market dynamics. Third, provide controlled empirical evidence that gradient boosting methods substantially outperform traditional regression and Random Forest when historical features are properly incorporated within a temporal validation framework. These contributions extend to other economic forecasting domains characterized by temporal autocorrelation and geographic heterogeneity.

2. DATASET AND METHODOLOGY

2.1 Dataset Description

The Redfin Housing Market dataset (Thuy, 2021) aggregates monthly housing market statistics at the county level across the United States from January 2012 through December 2021. This comprehensive dataset provides a 120-month time series encompassing both the recovery period from the 2008 financial crisis and the unprecedented COVID-19 pandemic market disruption. The dataset originally contained **563,122** observations representing unique combinations of county, property type, and month across 1,860 counties in 47 states.

Each record includes 58 features capturing various aspects of housing market activity. The target variable is median_sale_price, representing the median final sale price of all residential properties sold within a county during a given month. Median price was selected over mean price due to its robustness to extreme outliers common in real estate markets where luxury properties can substantially skew averages. Predictor variables span price metrics (median_list_price, median_ppsf with month-over-month and year-over-year changes), transaction volumes (homes_sold, pending_sales, new_listings), inventory measures (total inventory, months_of_supply), and market dynamics (median_dom for days on market, avg_sale_to_list ratio, sold_above_list percentage, price_drops frequency, off_market_in_two_weeks indicator).

2.2 Data Quality Issues and Remediation

Initial exploratory analysis revealed three critical quality issues requiring systematic remediation. First, extreme placeholder values specifically $999,999,999 and occasionally $1 appeared in price columns serving as sentinel values indicating missing data rather than genuine prices. These placeholders appeared in approximately 2.5 percent of observations but completely distorted statistical analyses. Most critically, the correlation between median_list_price and median_sale_price appeared near zero with Pearson coefficient of only 0.006 before cleaning when these variables should exhibit strong positive correlation as asking prices largely determine final sale prices.

Second, missing values were prevalent across numerous features, with some variables missing in over 50 percent of observations requiring strategic decisions about imputation versus exclusion. Third, statistical outliers existed both as legitimate geographic variation reflecting genuine market diversity and as data collection artifacts requiring careful distinction to preserve authentic heterogeneity while removing measurement errors.

Data preprocessing followed a systematic four-stage pipeline. Stage one addressed placeholder values through aggressive filtering: observations with median_sale_price exceeding $2,000,000 or below $10,000 were removed as implausible for county-level medians, effectively eliminating sentinel values while retaining genuinely expensive markets. Stage two applied similar filtering to median_list_price. Stage three employed Interquartile Range based outlier detection, removing observations beyond three times IQR from first and third quartiles. This conservative threshold was deliberately chosen to preserve legitimate geographic price variation while removing statistical anomalies.

Stage four addressed missing values hierarchically: rows with missing target variables were excluded (1,685 records), columns with over 50 percent missingness were dropped, forward-fill imputation was applied within each county's time series to maintain temporal continuity, and median imputation handled remaining gaps using county-specific medians. The complete preprocessing reduced the dataset from 563,122 to 505,729 records, a 10.2 percent reduction justified by dramatic quality improvements including correlation increase from 0.006 to 0.712 between list and sale prices.

2.3 Feature Engineering

Feature engineering transformed 58 original variables into 119 predictive features organized across six conceptual categories reflecting domain knowledge about housing market dynamics. Temporal features (14 created) capture cyclical and trend patterns: year, month, quarter, season indicators (is_spring, is_summer, is_fall, is_winter), cyclical encodings using sine and cosine transformations of month providing continuous seasonal representation, is_peak_season flagging the high-activity May-July period, is_covid_period for pandemic-era disruptions, and years_since_start measuring elapsed time from dataset beginning.

Lag features (18 created) capture recent price history through 1-month, 3-month, and 12-month historical values for median_sale_price, median_list_price, homes_sold, new_listings, inventory, and median_dom. The one-month lag captures immediate momentum, three-month lag provides smoothed recent trends, and twelve-month lag enables year-over-year seasonal comparisons. Rolling window features (20 created) calculate moving statistics over 3, 6, and 12 months:

rolling means smoothing short-term noise, rolling standard deviations quantifying volatility, price_momentum_3mo capturing acceleration, and price_trend_6mo measuring sustained directional movement.

Year-over-year growth features (10 created) quantify annual trends through percentage and absolute changes over 12-month periods for key variables. Geographic features (6 created) encode spatial hierarchies: state_median_price aggregating to state level, price_deviation_from_state measuring county position relative to state median, state_price_index normalized to 100 for cross-state comparison, county_price_rank_in_state and county_percentile_in_state quantifying within-state positioning, and state_encoded for categorical representation. Market condition features (11 created) capture supply-demand dynamics: supply_demand_ratio, market_heat_index, competition_level, and binary flags for fast markets, slow markets, low supply, and high supply conditions based on established industry thresholds.

2.4 Temporal Validation Methodology

Proper validation methodology represents a critical contribution. Housing price data exhibits strong temporal autocorrelation making random train-test splitting methodologically inappropriate. Under random splitting, a model trained on randomly selected 2020 observations and tested on other 2020 observations performs interpolation rather than genuine forecasting. More critically, random splitting enables data leakage where features calculated using information from the future contaminate training.

We implement strict temporal splitting mirroring real-world forecasting: training on 2012-2018 data (342,257 observations, 67.7 percent), validation on 2019 data (52,987 observations, 10.5

percent), and testing on 2021 data (57,123 observations, 11.3 percent). We deliberately skip 2020 to create clean separation between pre-pandemic training/validation and post-pandemic testing, avoiding COVID-19's unprecedented disruption in validation. This temporal structure ensures models never observe future data during training, providing honest performance estimates directly applicable to deployment. Under this framework, historical price lags become valid predictors representing genuinely past information, whereas under random splitting they would constitute leakage.

2.5 Feature Selection and Leakage Prevention

With temporal validation established, careful feature selection distinguishes legitimate predictors from features constituting data leakage. Excluded features include current period derivatives of the target variable (median_sale_price_mom, median_sale_price_yoy, median_ppsf variants) calculated FROM the target being predicted, features calculated using target values (price_list_diff, price_list_ratio), and problematic interaction features that dominated preliminary models obscuring interpretable base features.

Included features total 83 legitimate predictors carefully vetted for temporal validity. Historical sale price lags become valid under temporal splitting as they represent genuinely past information when forecasting future periods. State-level aggregations calculated from training data provide valid geographic context. Rolling averages computed from historical periods capture recent trends without future contamination. Market indicators represent current conditions observable before price realization. This careful selection balances predictive power with methodological integrity ensuring performance estimates transfer to deployment scenarios.

2.6 Model Selection and Training

We train and compare four regression models of increasing complexity. Linear Regression serves as baseline assuming simple additive relationships between features and target. Ridge Regression applies L2 regularization with alpha equals 10 to handle potential multicollinearity among the 83 features. Random Forest employs ensemble learning with 100 decision trees, max_depth of 20, min_samples_split of 10, capturing non-linear relationships through recursive partitioning. XGBoost represents state-of-the-art gradient boosting using 300 sequential trees with learning_rate of 0.05, max_depth of 10, subsample of 0.8, and colsample_bytree of 1.0. Each model trains exclusively on 2012-2018 data with hyperparameters validated on 2019 and final evaluation on 2021 test set. Performance uses RMSE measuring dollar error, R^2 indicating variance explained, and MAE providing robustness to outliers.

3. RESULTS

3.1 Model Performance Comparison

Table 1 presents comparative performance of all four models evaluated on 2021 test data.

Model	RMSE ($)	R^2	MAE ($)
Linear Regression	68,374	0.631	47,772
Ridge Regression	68,374	0.631	47,772
Random Forest	67,799	0.638	47,350
XGBoost	**4,574**	**0.9983**	**2,828**

Table 1. Model Performance on 2021 Test Data

Linear Regression and Ridge Regression achieve identical performance with RMSE of $68,374 and R^2 of 0.631, explaining 63.1 percent of price variance with mean absolute error of $47,772. The identical performance indicates multicollinearity among the 83 features is not a substantial limiting factor for linear approaches, as L2 regularization provides no meaningful improvement over unregularized ordinary least squares.

Random Forest demonstrates modest improvement with RMSE of $67,799 and R^2 of 0.638, representing a 0.8 percent reduction in RMSE and 0.7 percentage point increase in R^2. This marginal gain suggests that while non-linear relationships and feature interactions exist, the parallel tree-building approach with 100 trees captures them only partially. The similar performance clustering of these three methods with R^2 ranging 0.631 to 0.638 indicates they operate within a similar capability ceiling for this task.

XGBoost achieves transformational performance with RMSE of $4,574 and R^2 of 0.9983, representing 93.3 percent reduction in error and explaining 99.83 percent of variance. The mean absolute error of $2,828 indicates typical predictions deviate less than 2 percent from actual prices. This exceptional performance stems from XGBoost's sequential tree-building where each of 300 trees focuses on correcting residual errors from the previous ensemble, with conservative learning rate of 0.05 preventing overfitting while moderate tree depth of 10 balances complexity and generalization.

3.2 Achievement of Project Goals

The XGBoost model substantially exceeds both established project goals. RMSE of $4,574 surpasses the target threshold of below $25,000 by a factor of 5.5, achieving 81.7 percent better performance than required. This level of precision translates to average prediction errors between 1.1 and 3.0 percent depending on local price levels across markets typically ranging $150,000 to $400,000. Such precision proves sufficient for high-stakes decisions including investment capital allocation, mortgage underwriting, and policy planning.

The R^2 of 0.9983 substantially exceeds the target of 0.85, explaining 14.8 percentage points more variance than required and approaching the practical upper limit given inherent randomness in market transactions. In economic forecasting, R^2 values above 0.90 are uncommon due to unmeasured factors and behavioral unpredictability. The achievement of 0.9983 demonstrates that county-level median prices exhibit extremely strong predictability over short horizons when comprehensive features and advanced algorithms are properly applied with rigorous validation preventing optimistic bias from leakage.

3.3 Feature Importance Analysis

XGBoost feature importance scores quantify which predictors most strongly influence forecasts. The top five features collectively account for 44.2 percent of total importance: median_sale_price_lag3 (11.65%), median_sale_price_lag12 (9.82%), median_list_price_lag3 (8.94%), median_sale_price_lag1 (7.41%), and state_median_price (6.38%). This concentration among historical price features confirms housing markets exhibit strong momentum where recent price trends continue over short horizons, driven by both psychological factors including buyer anchoring to recent sales and structural factors including slow-changing housing stock characteristics.

The second tier includes market dynamics and supply indicators. Pending_sales_mom capturing month-over-month changes in properties under contract accounts for 4.23 percent, providing leading indicators of future sales. Supply_demand_ratio quantifying inventory relative to sales contributes 3.85 percent, identifying markets where tight supply supports appreciation versus markets with inventory overhang. Homes_sold_lag3 and inventory_lag3 add 4.10 and 3.65 percent respectively, capturing recent volumes and supply dynamics complementing primary price signals. Notably, simple interpretable features like historical lags outperform complex manually-engineered interactions.

3.4 Performance Consistency

Temporal consistency: Monthly RMSE across 2021 ranges from $3,890 (February, best month) to $5,420 (November, weakest month) with standard deviation of only $421. This narrow range demonstrates stable accuracy across different seasons and market conditions throughout a year characterized by historically unusual dynamics including unprecedented appreciation and intense competition. Slight degradation in late 2021 coincides with rapid acceleration, suggesting the model exhibits conservative bias slightly underpredicting during unusual rapid appreciation—a desirable property preventing encouraging overbidding based on overly optimistic forecasts.

Geographic consistency: RMSE varies from approximately $2,100 in stable Midwest markets to $8,400 in volatile coastal markets. However, this absolute variation largely reflects proportional consistency rather than differential quality. A $8,000 error on $600,000 median represents 1.3

percent, comparable to $3,000 on $200,000 representing 1.5 percent. This confirms the model successfully adapts to different price levels through state_median_price and state_price_index features normalizing for geographic cost differences.

Residual diagnostics confirm well-behaved errors. Mean residual of negative $127 indicates minimal systematic bias with predictions essentially unbiased. Distribution approximates normality with slight right-skew showing somewhat more overpredictions than underpredictions. Plotting residuals against predictions reveals homoscedastic errors where variance remains constant across the price spectrum, indicating consistent percentage accuracy from low-priced to high-priced counties. No systematic patterns emerge by state, month, or property type, confirming effective generalization across categorical dimensions.

4. DISCUSSION

4.1 Key Insights

Temporal validation is fundamental. Random train-test splitting allows future data leakage producing misleading metrics that fail in deployment. Temporal splitting (train 2012-2018, validate 2019, test 2021) ensures honest evaluation and enables principled incorporation of historical lags—the strongest predictors. This discipline prevented several methodological pitfalls including use of current period momentum and rolling averages calculated using forward-looking windows. The framework allowed valid incorporation of historical lags which proved critical, accounting for 44 percent of model importance.

Historical prices dominate predictive power. Top features are price lags at 1, 3, and 12 months collectively accounting for 44 percent of importance. Markets exhibit momentum over these horizons from both psychology (buyers anchor to recent comparable sales and extrapolate trends) and structure (housing stock and neighborhood characteristics change slowly). The twelve-month lag specifically captures year-over-year seasonal patterns enabling comparison of current prices to the same month one year prior. Simple extrapolation of recent trends, properly weighted, outperforms complex market indicators.

XGBoost's sequential learning achieves superior performance. The substantial gap between XGBoost R^2 of 0.998 and Random Forest R^2 of 0.638 demonstrates gradient boosting's fundamental advantages. While Random Forest builds independent trees in parallel from random subsamples, XGBoost builds trees sequentially where each focuses on correcting errors from the previous ensemble. This sequential error minimization through gradient descent in function space, combined with conservative learning rate and careful regularization, captures subtle patterns Random Forest misses. The 300-tree ensemble with learning rate of 0.05 requires substantially more training time but delivers transformational accuracy justifying computational investment.

4.2 Limitations

Short prediction horizon. Model accuracy applies specifically to 1-3 month forecasts. Longer horizons face structural shifts from interest rate changes affecting affordability, economic cycles influencing employment and income, demographic trends altering regional growth, policy interventions changing supply constraints, and technological disruptions enabling geographic arbitrage. Recursive forecasting compounds errors catastrophically. Model reliance on recent history assumes trends continue valid for weeks/months, increasingly tenuous over quarters/years.

Cannot predict unprecedented events. Training on 2012-2019 experienced exclusively recovery and expansion, never crashes or shocks. Model cannot reliably predict black swan events including financial crises, pandemic disruptions, or future shocks from policy changes, natural disasters, or technological disruptions. The test period (2021) occurred during unusual conditions yet performed well because patterns manifested as accelerated versions of 2019 validation trends. If asked to predict during sudden crashes with no training precedent, the model would fail. This limitation is inherent to all data-driven approaches: models interpolate within training experience but cannot reliably extrapolate to fundamentally different regimes.

Geographic aggregation. County medians aggregate substantial within-county heterogeneity. A $250,000 median might span properties from $150,000 to $500,000 depending on square footage, bedrooms, condition, specific neighborhood, school quality, and amenities. Individual property valuation requires property-specific features and different modeling approaches including spatial analysis of comparables. County-level predictions serve macroeconomic analysis, investment allocation, and policy planning rather than individual transactions.

4.3 Unexpected Challenges

Placeholder values ($999,999,999) completely distorted correlations—list-to-sale price appeared uncorrelated (r=0.006) when actually strong (r=0.712 post-cleaning). This issue appeared in only 2.5 percent of observations yet completely masked the fundamental relationship. Diagnosis consumed substantial time but taught a critical lesson: never assume public datasets are clean. Always validate basic relationships through scatter plots and correlation analysis before complex modeling.

Initial data leakage through features calculated FROM the target (median_sale_price_mom representing month-over-month sale price change, median_ppsf dividing sale price by square footage) produced unrealistically high preliminary performance (R^2=0.9992 under random splitting) that didn't generalize under temporal validation. Identifying these leaks required systematic auditing of every engineered feature determining whether it would genuinely be available at forecast time or incorporated information only observable after price realization. This process proved iterative as some features seemed legitimate until careful consideration revealed subtle dependencies on future information.

4.4 Practical Applications

Real estate investors can systematically screen all counties for favorable momentum before broader market recognition drives up valuations. By generating 1-3 month ahead forecasts across markets, investors identify opportunities where predictions substantially exceed current prices while fundamentals support the forecast. The $4,574 average error enables confident capital allocation distinguishing genuine opportunities from statistical noise.

Municipal governments can use forecasts to anticipate affordability pressures and plan proactive interventions. If models predict rapid growth in affordable counties, policymakers can expand housing supply, adjust zoning, or implement targeted measures before crises emerge. Property tax revenue forecasting improves with accurate price predictions, enabling better budgeting for schools and infrastructure funded through property taxation.

Financial institutions can incorporate forecasts into mortgage risk models recognizing predicted appreciation reduces default risk as borrower equity strengthens. Lenders might offer marginally lower rates in high-growth-forecast markets reflecting reduced risk or require larger down payments in weak-forecast markets. Mortgage servicers can identify geographic concentrations in adverse-forecast markets enabling proactive loss mitigation. Real estate investment trusts can optimize acquisition and disposition timing across markets based on expected trajectories.

5. CONCLUSIONS

5.1 Summary

This research successfully developed machine learning models for county-level median housing price prediction, substantially exceeding both performance targets with RMSE of $4,574 and R^2 of 0.9983. These results demonstrate that short-term forecasting (1-3 months) achieves exceptional accuracy when combining proper temporal validation preventing leakage, comprehensive feature engineering incorporating historical lags and market dynamics, and gradient boosting methods learning complex interactions. Historical price lags account for 44 percent of predictive power confirming market momentum dominates short-term dynamics. The analysis establishes both the power of data-driven approaches and their inherent limitations for unprecedented events.

5.2 Contributions

This work establishes empirical benchmarks showing errors below $5,000 (approximately 2.5 percent of median prices) are attainable with proper data quality management, comprehensive feature engineering, and advanced algorithms. It provides replicable feature engineering frameworks specifying which temporal lags, rolling statistics, and market indicators drive accuracy. It demonstrates temporal validation's critical importance ensuring honest performance assessment for time-series forecasting. It validates gradient boosting's substantial superiority over traditional regression and ensemble methods for structured data with complex interactions.

5.3 Future Research Directions

Future work should incorporate macroeconomic indicators including interest rates, unemployment, wage growth, and GDP enabling longer-term forecasting and structural shift detection. Extending to property-level prediction would add property-specific features including square footage, bedrooms, condition, and precise location supporting individual home valuation for refinancing and negotiations. Developing uncertainty quantification through prediction intervals via Bayesian methods or conformal prediction would enable risk-adjusted decision-making with explicit probability bounds. Applying explainability techniques specifically SHAP values would decompose predictions into feature contributions building user trust. Building production deployment infrastructure with continuous data pipelines, automated retraining, performance monitoring, and drift detection would transform academic research into operational decision support systems.

5.4 Final Remarks

The performance achieved demonstrates that county-level housing price forecasting for 1-3 month horizons represents a solved problem given sufficient data, appropriate methodology, and advanced algorithms. This should embolden practitioners to deploy similar models operationally while acknowledging that accuracy degrades for longer horizons, unprecedented market conditions, and individual property valuations requiring additional context. The techniques employed—gradient boosting, comprehensive feature engineering, temporal validation—represent current best practices transferable to other time-series forecasting domains in economics, finance, and engineering exhibiting similar patterns of temporal autocorrelation and geographic heterogeneity. As housing markets continue evolving with new technologies, remote work enabling geographic arbitrage, and climate change affecting regional desirability, sophisticated data-driven forecasting becomes increasingly valuable for efficient resource allocation across the housing economy.

REFERENCES

Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5-32. https://doi.org/10.1023/A:1010933404324

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (pp. 785-794). https://doi.org/10.1145/2939672.2939785

Géron, A. (2019). Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow (2nd ed.). O'Reilly Media.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825-2830.

Thuy, L. (2021). Redfin housing market data [Data set]. Kaggle. https://www.kaggle.com/datasets/thuynyle/redfin-housing-market-data

Zillow Research. (2021). Total value of all US homes. Retrieved from https://www.zillow.com/research/

AI USE DISCLOSURE

In accordance with Northeastern University's Academic Integrity policy, AI tools were utilized during this project for conducting background research on machine learning methodologies, debugging Python code when encountering technical errors, identifying and resolving data quality issues including placeholder value detection, and building a structured roadmap for the project workflow. All conceptual work, analytical decisions, feature engineering strategies, model selection rationale, results interpretation, and written content represent myself (Akash Ghosh) original contributions. All code was personally executed by me, and all results presented are based on actual analysis I conducted.